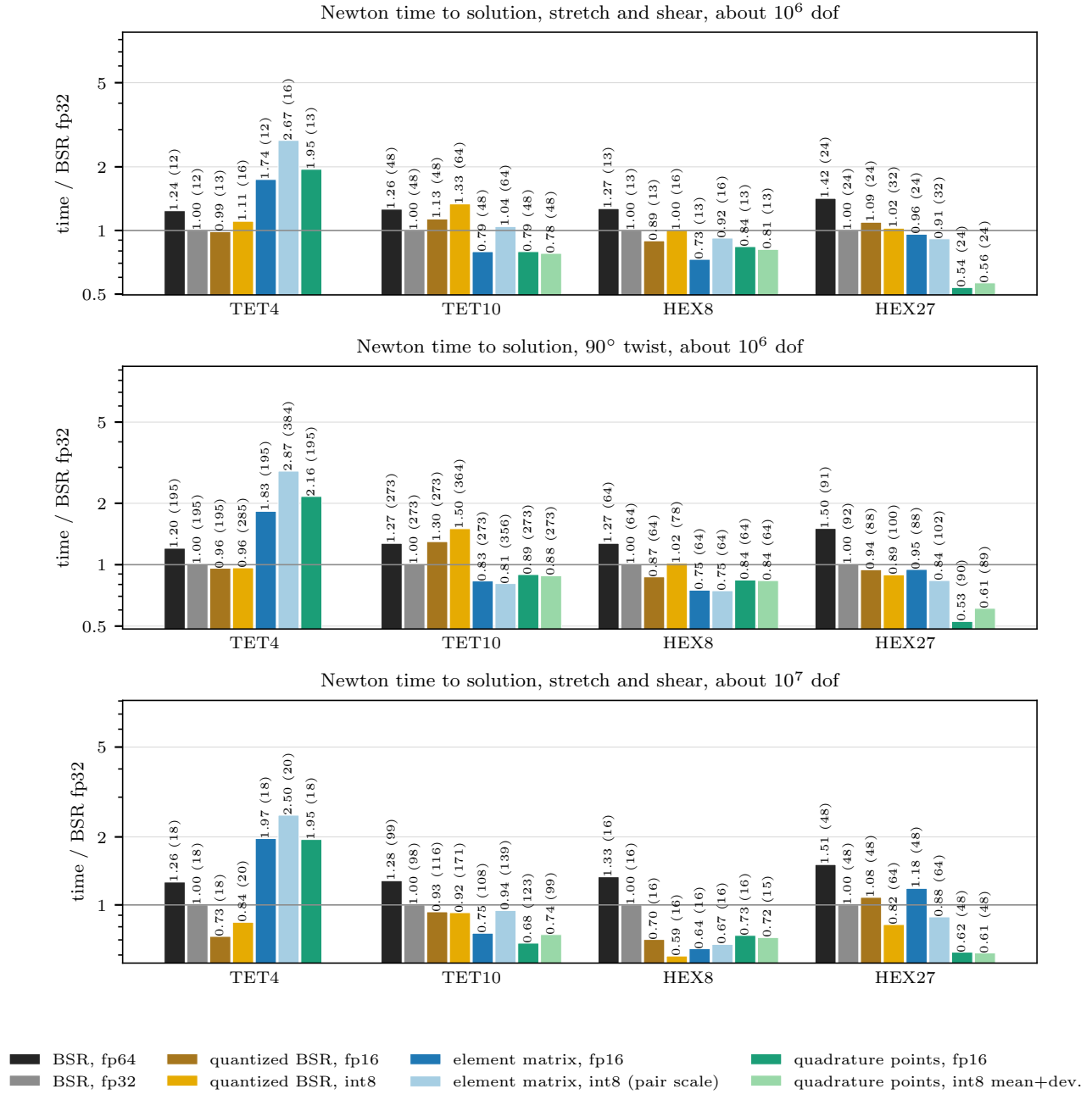

Newton time to solution, stretch and shear, about 10^6 dof
time / BSR fp32
0.5
1
2
5
1.24 (12)
1.00 (12)
0.99 (13)
1.11 (16)
1.74 (12)
2.67 (16)
1.95 (13)
1.26 (48)
1.00 (48)
1.13 (48)
1.33 (64)
0.79 (48)
1.04 (64)
0.79 (48)
0.78 (48)
1.27 (13)
1.00 (13)
0.89 (13)
1.00 (16)
0.73 (13)
0.92 (16)
0.84 (13)
0.81 (13)
1.42 (24)
1.00 (24)
1.09 (24)
1.02 (32)
0.96 (24)
0.91 (32)
0.54 (24)
0.56 (24)
TET4
TET10
HEX8
HEX27
Newton time to solution, 90° twist, about 10^6 dof
time / BSR fp32
0.5
1
2
5
1.20 (195)
1.00 (195)
0.96 (195)
0.96 (285)
1.83 (195)
2.87 (384)
2.16 (195)
1.27 (273)
1.00 (273)
1.30 (273)
1.50 (364)
0.83 (273)
0.81 (356)
0.89 (273)
0.88 (273)
1.27 (64)
1.00 (64)
0.87 (64)
1.02 (78)
0.75 (64)
0.75 (64)
0.84 (64)
0.84 (64)
1.50 (91)
1.00 (92)
0.94 (88)
0.89 (100)
0.95 (88)
0.84 (102)
0.53 (90)
0.61 (89)
TET4
TET10
HEX8
HEX27
Newton time to solution, stretch and shear, about 10^7 dof
time / BSR fp32
1
2
5
1.26 (18)
1.00 (18)
0.73 (18)
0.84 (20)
1.97 (18)
2.50 (20)
1.95 (18)
1.28 (99)
1.00 (98)
0.93 (116)
0.92 (171)
0.75 (108)
0.94 (139)
0.68 (123)
0.74 (99)
1.33 (16)
1.00 (16)
0.70 (16)
0.59 (16)
0.64 (16)
0.67 (16)
0.73 (16)
0.72 (15)
1.51 (48)
1.00 (48)
1.08 (48)
0.82 (64)
1.18 (48)
0.88 (64)
0.62 (48)
0.61 (48)
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.